My introduction to teaching computer science concepts began during an undergraduate summer research program at Princeton University, when I taught a lecture on approximation algorithms[1] while the instructor was traveling. It took me a whole week, and many practice runs, but the process of preparing to teach complex concepts in a creative way, showed me how much care and effort is required to effectively teach, and also how much I enjoyed it. Since then, I have taken advantage of different teaching opportunities during my PhD, and as an industry researcher through:

- **Running a summer course**: In the final year of my PhD, I ran an undergraduate research summer course at Brown. This involved selecting applicants from a large candidate pool, designing course materials and TAs duties, interacting with administrators for room assignments and bookings, holding office hours, and preparing materials for lectures, assignments, and research programs.
- **Summer schools, guest lectures & seminars** I've been invited to guest lecture on specific course materials at Brown University, UC Berkeley, Pomona University, Rutgers University, Princeton University. I also taught a series of safety and language modeling lectures at the M2L summer school in 2022 and 2024.
- **Advising students & teaching researchers**: As an industry researcher, I've advised 6 PhD students during internships at DeepMind. I mentored several undergrads, as well as younger grad students during my PhD. I taught tutorials to non-language researchers at DeepMind: when the company pivoted towards LLM products, I was asked to do a 1-week tutorial on training language models for the multi-agent and game theory team.
- **Advisory boards & academic engagement**: I was invited to be part of an academic advisory board at Brown University, to help shape future CS courses that incorporate, rather than ban, the use of AI in education. I have also co-taught tutorials on language models to academic communities.

My philosophy towards teaching and mentorship has been shaped by the above experiences. Through these I have developed a teaching philosophy centered on three principles: encouraging critical thinking, building intuition through concrete examples, and being able to adapt to each student. As a professor, I hope to bring these principles to both classroom instruction and research advising.

Teaching & Lecturing

I strongly believe that students learn best when encouraged to do two things: first, to focus on the concepts, and learn through concrete examples that illustrate complex methods; and to explicitly question the assumptions, limitations and mechanisms behind them. This is even more crucial in this era of AI assistant tools whose help, when used incorrectly, might actually hinder, rather than aid student's learning and developmental processes.

Through my career, both graduate school and industry, I've had the opportunity to teach large scale tutorials attended by upto 1000 virtual participants, summer school lectures attended by upto 300 in-person students, college-level seminars attended by small groups of graduate students, and industry tutorials attended by hundreds of post-graduate researchers across disciplines. These audiences have varied in their backgrounds, age groups, disciplines, and research interests; requiring a balance of teaching conceptual intuition along with necessary implementation details, while ensuring that every person's academic needs are fulfilled.

Moreover, being on the academic advisory board at Brown University has furthered my motivation to be a part of the academic teaching ecosystem. In this era of chatbot assistants that make traditional teaching and homeworks hard, I'm excited to find new ways to challenge and evaluate students. I think this is a hard problem, but a necessary, and even exciting one for us to tackle: on how to educate this new wave of young minds that grow up with chatbot assistants, to ensure that the fundamental development processes that college educations aim to instill in them, still function in the same way as they did for previous generations.

[1] We had been doing research on this topic that whole summer. The semester after, I discovered research in Natural Language Processing and never went back to theoretical computer science research.

Mentoring & Advising

The most fulfilling part of my job at DeepMind has been mentoring. As a senior researcher, I've been fortunate to have 6 PhD students either directly or co-advised by me through various internships. These collaborations have led to first-author research papers by the students I have mentored; but more importantly, have shaped my understanding of what effective research advising requires. I've also been an active part of Google's internal career mentorship program, having mentored 11 engineers and researchers on career development and industry jobs.

My mentoring philosophy is to provide substantial structure early in a project while gradually increasing students' intellectual ownership. At the beginning of a collaboration, I work closely with students to identify a tractable research question, establish meaningful baselines, and develop a concrete experimental plan. As the project progresses, I encourage students to propose alternative hypotheses, make methodological decisions, and critically evaluate their own findings.

[For example, one intern initially approached a research question with a broad set of possible experiments but struggled to identify which would provide the most convincing evidence. In our early meetings, we worked together to isolate the central hypothesis and design a small set of decisive experiments. As the results emerged, I encouraged the student to lead discussions of their interpretation and propose the next steps. By the end of the internship, the student had developed the confidence to independently defend the project's methodological choices and lead the writing of a first-author paper.]

Experiences like this have taught me that successful mentorship requires more than technical guidance. Students must also learn how to recognize important research questions, navigate uncertainty, respond constructively to criticism, and communicate their ideas clearly. I therefore devote time to discussing experimental design, scientific writing, presentation skills, and the broader significance of research results. I encourage students to see unsuccessful experiments not simply as setbacks, but as opportunities to refine their understanding of the problem.

Adapting mentorship to individual students. Students enter research with different strengths, interests, and levels of experience. Some are confident implementers who need support in identifying research questions; others have strong conceptual intuitions but require guidance in experimental methodology. I aim to tailor my mentoring accordingly, balancing regular individual meetings with opportunities for students to develop independence.

I also believe that students should feel comfortable disagreeing with their advisors. Research advances when assumptions are challenged, and junior researchers often bring perspectives that more experienced collaborators overlook. I try to create an environment where students can question my suggestions, propose alternative directions, and take ownership of ideas that differ from my initial expectations.

As a faculty member, I plan to build a research group in which students receive individualized guidance while also learning from one another. I envision regular individual meetings, collaborative research discussions, paper-reading groups, and opportunities for senior students to mentor junior researchers. My goal is to help students develop not only strong technical skills, but also the independence, judgment, and confidence needed to become successful researchers and collaborators.

I also co-organized a tutorial at the ACL 2021 virtual conference **?**. This ACL tutorial was attended by more than 1000 attendees virtually.

Future Teaching

I'm excited to teach both types of classes: more general, introductory computer science classes (e.g., artificial intelligence or machine learning) and in-depth, graduate level seminars on emerging topics (e.g., modeling safety risks or longitudinal data collection).

I am particularly enthusiastic about teaching computer science courses that my research and expertise span i.e., natural language processing, reinforcement learning, deep learning, safety and ethics, alignment. Separately, I'm very excited to formulate interdisciplinary courses e.g., examining the relationship between AI and society, human-

AI interactions, large-scale data collections, social impacts and foresight about increasingly capable AI systems. Mentoring and teaching have been such a fundamental part of my motivation to continue an academic career, and I look forward to extending this experience to the classroom and building a supportive research environment in which students from diverse backgrounds can develop the skills, curiosity, and confidence to shape the future of artificial intelligence.